Antibiotics and Antivirals



Specific Resistance to Infection

Antibiotics (antimicrobials)

- Not the same thing as antibodies, antigens, or vaccines!
- Anti – “against” Biotics – “living things”
- Class of drugs used to fight micro-organisms – usually bacteria
- Generally not preventative – they are given *after infection* to help kill the pathogen
- Revolution in treatment of infection
  - Before penicillin, many deaths resulted
- Many classes of antibiotics, depending on mode of action
  - Penicillins: prevent bacteria from developing cell walls, therefore inhibiting reproduction
  - Actinomycins: interfere with protein synthesis in bacterial cells
  - Cephalosporins: interfere with bacterial cell wall synthesis



Learning Aim: Define the term “antibiotics”

Antibiotics

- Two broad types:
  - **Bactericidal – kill bacteria** by:
    - Inhibiting cell wall synthesis
    - Inhibiting bacterial enzymes
    - Inhibiting protein translation within the bacterium
  - **Bacteriostatic - prevent bacteria from multiplying** while the immune system deals with them.
    - Inhibit bacterial protein production
    - Inhibiting bacterial DNA replication
    - Interfering with other aspects of bacterial metabolism

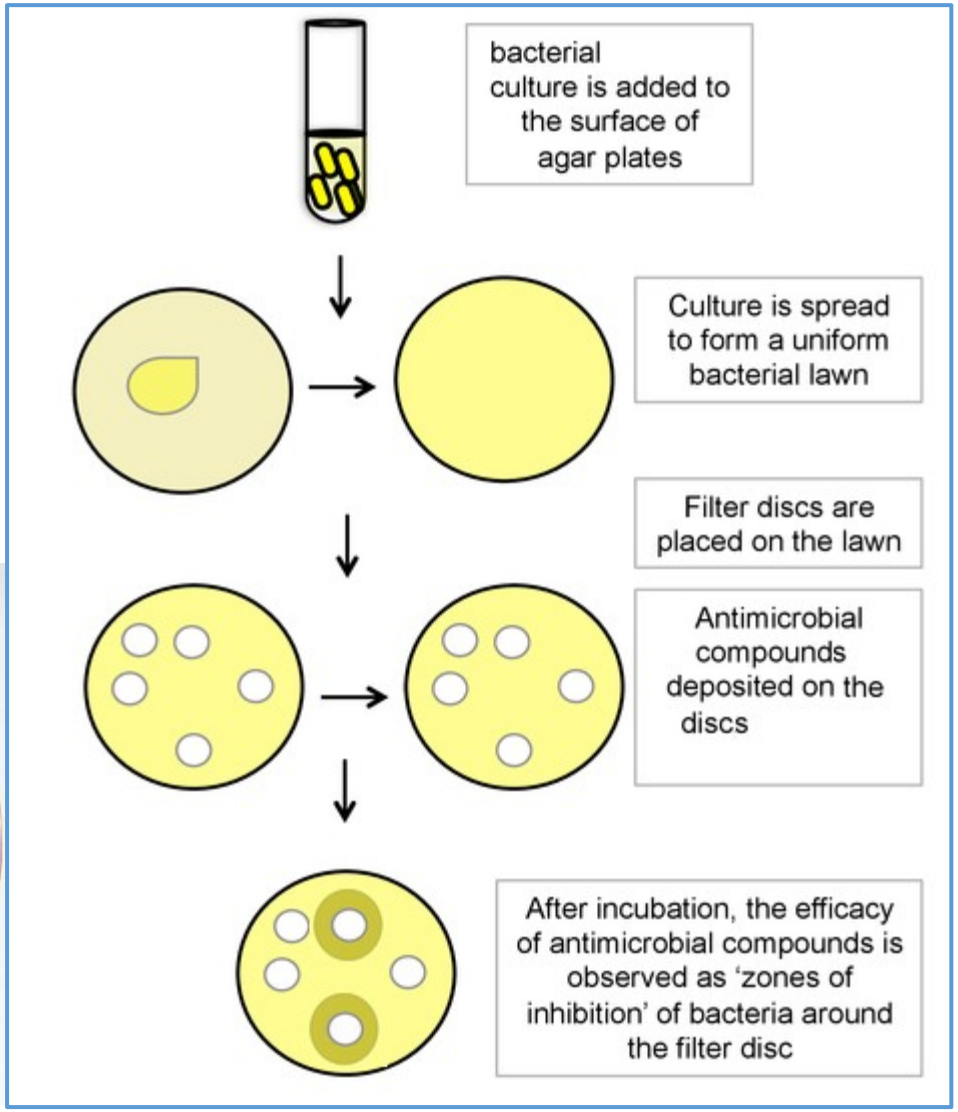
- Different spectrums of action:
  - **Broad spectrum:**
    - Work against a wide range of bacterial types
  - **Narrow-spectrum**
    - Effective against a narrow range

Learning Aim: Describe the two broad types of antibiotics and how each works | Describe the spectrums of action of antibiotics

Antibiotics

- Each type of antibiotic only works against certain types of bacteria, even if broad spectrum.
  - Culture and sensitivity is used to determine which bacterium is present and which antibiotic will work.
    - Bacteria smeared on petri dish
    - Discs with antibiotic put on
    - Bacteria don't colonise areas with antibiotic that kill that bacterium





- Antibiotics DO NOT work on viruses
- Prescription from Dr required

Learning Aim: Describe how culture and sensitivity works to determine an antibiotic that will work for a specific pathogen

Antibiotic Resistance

- Bacteria evolve over time, to become resistant to antibiotics.

- New antibiotics must be developed to combat this.

- Things that contribute to resistance:
  - Overuse of antibiotics, even when not necessary.
  - Agricultural use as ‘growth promoters’ in livestock.
  - Incorrect prescribing: giving an antibiotic that won’t work, or as a preventative without good reason.

- Some bacteria now show multiple drug resistance and total drug resistance, meaning we no longer have treatment for them.

Learning Aim: Explain why antibiotic resistance is a concern and outline things that contribute to antibiotic resistance

Antivirals

- Specifically for viral infections.

- Difficult to find drugs for this, as virus enters cell – any drug that interferes with virus may also harm host cell.

- Research today targets specific viral proteins (not found in humans) to disable the virus.

- Antivirals inhibit development of virus, rather than killing it.
  - “Zovirax” (acyclovir) – herpes
  - Interferons – Hepatitis B
  - AZT (azidothymidine) - HIV
  - “Tamiflu” (Oseltamivir) - influenza



Learning Aim: Describe how antiviral drugs work, give examples of antiviral drugs and the diseases they treat